Multimodal Fusion for Objective Assessment of Cognitive Workload: A Review

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***Abstract*—Considerable progress has been made in improving the estimation accuracy of cognitive workload using various sensor technologies. However, the overall performance of different algorithms and methods remain suboptimal in real-world applications. Some studies in the literature demonstrate that a single modality is sufficient to estimate cognitive workload. These studies are limited to controlled settings, a scenario that is significantly different from the real world where data gets corrupted, interrupted, and delayed. In such situations, the use of multiple modalities is needed. Multimodal fusion approaches have been successful in other domains, such as wireless-sensor networks, in addressing single-sensor weaknesses and improving information quality/accuracy. These approaches are inherently more reliable when a data source is lost. In the cognitive workload literature, sensors, such as electroencephalography (EEG), electrocardiography (ECG), and eye tracking, have shown success in estimating the aspects of cognitive workload. Multimodal approaches that combine data from several sensors together can be more robust for real-time measurement of cognitive workload. In this article, we review the published studies related to multimodal data fusion to estimate the cognitive workload and synthesize their main findings. We identify the opportunities for designing better multimodal fusion systems for cognitive workload modeling.**

***Index Terms*—Cognitive workload, decision fusion, information fusion, mental load, multimodal data fusion, task load.**

I. Introduction

THE HUMAN operator is a central part of any form of human–automation interaction system. Cognitive workload of human operators affects the design and implementation of human–robot systems. By evaluating the operator cognitive workload, problems, such as work overload and underload, can be identified and mitigated to improve the efficiency of a system.

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The term *Cognitive workload* has been defined in different ways in the literature. Wickens defined cognitive workload as "relation between the (quantitative) demands for resources imposed by a task and the ability to supply those resources by the operator" ([1, p. 161]). Cain defined it as "a mental construct that reflects the mental strain resulting from performing a task under specific environmental and operational conditions, coupled with the capability of the operator to respond to those demands" ([2, p. 4]). Cognitive workload has been explained in [3] as a concept that is identified as observed delays in information processing capabilities when a considerable amount of mental effort is exerted by an individual. It has also been explained in [4] as a phenomena that happens when the required capacity of the information-processing system exceeds the available capacity.

Abbass *et al.* [5] distinguished the different concepts used in the literature, where they distinguished between the mental load related to the work environment (workload) and that imposed by the exogenous environmental factors such as an operator's personal life (environmental load). The interaction interface was added to the equation to clearly isolate the effect of the interface from the main task. The two equations used by Abbass *et al.* [5] are:

$$\mathrm{CL} = \mathrm{WL} + \mathrm{EL}$$
$$\mathrm{WL} = \mathrm{TL} + \mathrm{IL} + \mathrm{OF}$$

where CL stands for cognitive load, WL for workload (mental workload in this article), TL for task load (task workload in this article), IL for interface load (not considered in this article), EL for environmental load, and OF for other work related factors. Accordingly, they defined CL as "the amount of mental resources needed to perform the task" ([5, p. 9]). We will rewrite the above equations as follows:

$$\mathrm{CL} = \mathrm{MWL} + \mathrm{EL}$$
$$\mathrm{MWL} = \mathrm{TWL} + \mathrm{DF}$$

where MWL = WL and DF = IL + OF, where MWL refers to mental workload, TWL denotes task workload, and DF denotes the disturbances from factors other than the task itself.

It is worth noting that the only component that is normally under the control of a task designer is the task workload component, which is the focus of this article. The remaining components in the theory are either compound factors that are uncontrollable, which include the environment load and other covariates representing factors that are unknown to the analyst,

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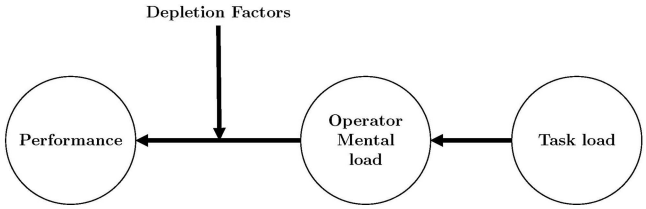


Fig. 1. Cognitive workload components.

or factors caused by the interface that are unrelated to the task but are controllable through other means. Therefore, this article only covers TWL, which is the primary controllable factor in the hand of a task designer.

Among cognitive workload definitions and explanations proposed in the literature, the definitions proposed by Wickens [1], Cain [2], and Abbass *et al.* [5] explain the different components and external factors of cognitive workload. To summarize these three definitions, cognitive workload can be defined as a multidimensional concept that consists of four components: 1) task complexity; 2) mental workload; 3) performance; and 4) depletion factors. Task complexity characterizes the difficulty of a given task and is an inherent property of the task independent of the human involved in the task. Mental workload refers to the level of mental resources an individual is able to supply to maintain a high-performance level while performing the task. The performance component represents the interaction between task complexity and operator mental workload. Depletion factors are external factors affecting the operator's mental capacity, such as stress, fatigue, motivation, task importance, and attitude (see Fig. 1).

This article aims to summarize the studies presented in the literature that attempted to use multimodal physiological data to predict TWL imposed on human operators.

Adopting such multidimensional decomposition of cognitive workload concept simplifies the taxonomy of the previous work in the cognitive workload literature. It highlights the dynamic relationships among different components. Among the four components of cognitive workload, our focus will be the workload and depletion factor components.

Subjective cognitive workload commonly refers to the personal perception of self-assessment of workload. It normally gets measured using surveys or questionnaires [6]. Objective workload focuses on identifying workload levels using reproducible and quantitative data, such as physiological or neurophysiological responses of human operators to certain task demands [7], [8].

In general, most objective measures can be classified either as neurophysiological, physiological, or behavioral. Neurophysiological measures include EEG and functional near-infrared spectroscopy (fNIRS) [9]. Physiological measures include electrocardiography [10] and eye tracking (pupil dilation [11], blink frequency and blink duration [12], and saccades [13]). Behavioral measures include keystroke dynamics, mouse tracking, and body positioning [14]. However, the accuracy, robustness, temporal resolution, and latency of these approaches are suboptimal and, in certain contexts, do not respond to changes in task load at all [3], [15]. This is due to the measurement noise, artifacts induced by a subject's movements, and the limited amount of training data available. Developing a robust and accurate metric that can quickly and efficiently reflect the cognitive workload remains a challenging task.

In recent years, novel strategies have been explored to overcome these limitations. One such approach is multimodal data fusion; a group of techniques that allow incorporation of data from different sources (from the same or different modalities) to infer important information that is not accessible by using a single independent source [16], [17]. The concept of data fusion from different sources is not new, indeed, humans and animals use multiple senses (e.g., vision, smell, and hearing) to obtain valuable information and to be able to assess different situations for survival. For instance, in situations with low visibility, such as in darkness or dense fog, the senses of touch and hearing can be used to overcome visual limitations. Therefore, a combination of different senses helps to construct a more accurate assessment of a given situation. Similarly, by utilizing multiple measures together to assess workload, robust and enhanced results can be achieved [18]–[21].

In this article, the application of multimodal data fusion approaches to cognitive workload modeling is discussed. In particular, this article aims to highlight the importance of multimodal fusion frameworks to account for specific properties of each data source that could better explain the workload experience by the subject. The remainder of this article is organized as follows. Section II presents a summary of common approaches for assessing cognitive workload. Section III exhibits the limitations of using single measures for prediction of workload. Section IV offers a review of multimodal data fusion approaches in cognitive workload modeling. Section V presents the design considerations for effective workload estimation. Finally, the conclusions are given in Section VI.

II. Common Objective Approaches for Assessing Cognitive Workload

Before discussing the multimodal fusion literature of workload, this section summarizes some of the popular approaches for the objective assessment of cognitive workload.

A. Neurophysiological Measures

Neurophysiological activity is perhaps the most direct indicator of cognitive workload. There are two main techniques to measure cognitive workload using brain activity: 1) EEG and 2) fNIRS.

EEG measures the brain's electrical activity and the pattern analysis of this activity is used to indicate different levels of cognitive workload. Spectral analysis decomposes EEG signals into its constituting frequency components. Typically, EEG data are partitioned into five bands (from slowest to fastest: delta, theta, alpha, beta, and gamma). The power spectral density (PSD) in each band is computed and used to compare the conditions being studied (i.e., low versus high workload). EEG is considered the most popular approach in the literature to objectively assess cognitive states [22]–[25]. Another

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approach to assess cognitive workload is to measure activation of different brain regions by examining the levels of oxygenation (HbO) and deoxygenated (HbR) hemoglobin concentration in the cerebral cortex [26]. fNIRS is commonly used in this regard to measure the amount of effort exerted in a given brain region in response to a given task. Different studies have reported that increased levels of HbO in the prefrontal cortex correlates with increased task engagement which is used to indicate the increased cognitive workload [27], [28].

B. Physiological Measures

In neuroscience studies, mental workload is linked to changes in the activity of the autonomic nervous system (ANS) [10], [29]. Mental stressors impact involuntary activities such as those induced by cardiac and respiratory activities.

Under high workload conditions, heart rate rises to supply the body with elevated levels of oxygenated blood. ECG measures the electrical activity generated by the heart and provides an indicator of the ANS and stress states. Metrics, such as heart rate variability (HRV), interbeat-interval (IBI), and blood pressure (BP), have been shown to be sensitive to increased cognitive effort [30]. Increased task difficulty was shown to be strongly correlated with decreased IBI and increased BP [10].

Respiration has also been considered as one of the most sensitive metrics of the mental task demands [31]. It has been reported that the human's respiratory cycle increases when completing complex tasks or during periods of stress, this phenomenon is called hyperventilation [32]. Respiratory volume tends to increase as workload increases [10]. The main respiratory metrics used to measure mental workload are: respiration rate, respiration variability, and respiratory airflow and volume [31].

Electrodermal activity (EDA) also known as galvanic skin response (GSR), in this method, the changes in electrical conductance of the skin in a person is measured through the use of sensors attached to the skin [33]. Skin conductance varies as sweat glands activate as an unconscious body function of the ANS [34]. In stressful conditions, the sympathetic nervous system and the parasympathetic nervous system (both part of ANS) are activated simultaneously but with an opposite effect. However, the skin glands and the blood vessels in the skin are only elicited by the sympathetic nervous system [35], in contrast with other physiological measures (i.e., cardiac activity) that are affected by parasympathetic activity [34]. In general, skin conductance shows a positive correlation with increase of workload [33].

Oculomotor measures are also a useful physiological measure for assessing cognitive workload [36]–[38]. One popular technique is electrooculography (EOG), this technique is used to record eye movements by measuring the electric potential field between the cornea and the retina, the potential difference is also called the corneoretinal potential (CRP) [39]. To measure the CRP of a subject, a pair of electrodes is placed above and below each eye, and eye movements are recorded as changes in the electric field [40]. Another popular technique to measure eye activity is pupillometry using cameras. Pupillometry is accomplished by measuring the pupil's response to different visual stimuli. Eye activity metrics, such as pupil dilation, blink frequency, and blink duration, correlate with cognitive workload with varying levels [41]–[43]. Previous studies have suggested that blink frequency and blink duration decline when task demands increase and during periods of high cognitive load [44].

C. Behavioral Measures

Behavioral analysis is an another useful approach for assessing cognitive workload. Behavioral metrics, including keystroke dynamics, mouse tracking, and body positioning, can be used as the effective metrics for assessing cognitive workload. In human–computer interaction (HCI), user attention and engagement can be measured by activities, such as clicking, scrolling, or keyboard typing [45]. For example, keystroke features (i.e., use of backspace, delete, end, and arrow keys) have shown changes in typing patterns to be affected by increased cognitive workload during mental multiplication and memory, which in turn affect the task performance [46]. In another study [47], mouse pressure was successfully used to measure workload with 88% accuracy. In that study, a computer mouse equipped with eight pressure sensors was used to collect pressure signals while participants performed a computer task; it was found that mouse pressure increases with increased levels of frustration.

Body posture is another useful behavioral measure, which is attracting more research in HCI. Seated posture has shown to correlate with task engagement and boredom, increased pressure on the seat was associated with increased engagement, while increased pressure on the back of the chair was associated with boredom [48]. In another study, body posture was shown to be associated with stress levels and situational awareness [49]. A clear disadvantage of motor actions as a measure of cognitive workload is that task-performing body movements are less frequent (episodic) than other measures (i.e., physiological) [50].

III. Motivations for Multimodal Data Fusion for Cognitive Workload and Types of Fusion Systems

Due to the complex nature of cognitive workload, it is rare that a single measure provides a robust and/or comprehensive index of workload. Assessing cognitive workload based on a single sensor suffers several limitations.

1) *Reliability:* A sensor failure causes loss of the desired physical measure for workload. In a real-life application, this would eventually cause system failure.
2) *Uncertainty:* Sensors are generally affected by different sources of noise (e.g., instrumental and environmental), which makes them more sensitive to error. For instance, in measuring the cognitive workload by fNIRS, several confounds could affect the measure, such as skin and skull haemodynamics or motion artifacts [51], [52]. In this case, the uncertainty existing in the measurements will restrict the analysis.
3) *Sensitivity:* It is the responsiveness to different levels in task demands that are imposed on a person. Most workload assessment techniques are sensitive to certain types of workload. Therefore, looking at only one dimension at a time limits the predictability of the

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assessment. Ideally, a cognitive workload assessment should exhibit variations to different factors affecting the workload; Wierwille and Eggemeier name this as global sensitivity [53].

4) *Individual Variability:* Recent studies suggest that physiological responses to different workload measurements vary significantly from subject to another [54]. Therefore, depending on a single sensor to estimate workload does not scale well over a large population of subjects.

Multisensor fusion provides an effective solution to the aforementioned limitations. The fusion of data from multiple sensors introduces several advantages [55]–[58].

1) *Improved Signal-to-Noise Ratio:* Fusion of data streams from independent sensors reduce the noise effect.
2) *Enhanced Robustness and Reliability:* The redundancy resulting from the use of multiple sensors [i.e., measuring the heart rate using both electrocardiogram (ECG) and photoplethysmogram (PPG) sensors] enhance fault-tolerance of the system against sensor failure and reduce the bias of confounding factors.
3) *Reduced Ambiguity and Uncertainty:* Using data from multiple sources reduces uncertainty of the system output.
4) *Improved Resolution and Hypothesis Discrimination:* When multiple independent measurements of the same phenomena are fused, the resolution of the resulting value is higher than what can be achieved using a single sensor.
5) *Enhanced Performance:* Different sensors provide measurements for different aspects of studied phenomena. The fusion of these measurements produce higher level value than what can be achieved using a single sensor.
6) *Enhanced Temporal and Spatial Coverage:* Some sensors have high temporal but low spatial resolution while other sensors have higher spatial but lower temporal resolutions. Sensor fusion provides better coverage for both temporal and spatial resolutions. For example, EEG has high temporal resolution but low spatial resolution while fNIRS has high spatial but low temporal resolution. Combining both sensors provide rich information about the studied phenomena.

There are three common configurations for multimodal data fusion in [59] and [60].

1) *Competitive:* Different modalities provide independent measurement of the same aspect about a certain phenomenon. A fusion module then decides which modality correctly represent that phenomena. This type of fusion is used for fault-tolerance and robust systems because it can alleviate problems, such as noise and artifacts.
2) *Complementary:* Different modalities provide measurement (features) for different aspects of a certain phenomenon. Combining these measurements together provide a more complete understanding of the phenomenon under observation. This type of fusion is used to resolve the incompleteness of a single sensor data.
3) *Cooperative:* Data from different modalities are combined together in such a way that measurement can be derived that would be otherwise unavailable from single sensors.

In terms of data processing, multimodal fusion is typically implemented at three different levels of abstractions.

1) *Sensor Level:* Fusion module processes raw data captured from different sources. If multiple sensors are measuring the same physical aspect, a single feature vector to represent the phenomena under analysis can be directly combined. However, if sensor data represents different phenomena, the data fusion should be completed in a later stage.
2) *Feature Level:* Sensor data is represented by feature vectors. Features are extracted from different sources independently. The fusion module then combines the feature vectors from each module into a single-fused feature vector.
3) *Decision Level:* Features are extracted from each source independently and passed to a corresponding classification module to make their own decision. The fusion module then consolidates these decisions into one final classification decision.
4) *Hybrid Models:* These models can include more than one level of fusion. For example, features from two modalities can be combined together to construct one feature set for classification model, the decisions of this classification model are then combined with decisions from a second classification model that is trained using features from a third modality.

Each data fusion paradigm can be implemented at different level of processing abstractions. For example, competitive fusion can be implemented at the sensor, feature, or decision level. Fig. 2 presents the three types of fusion levels.

IV. Multimodal Data Fusion Approach to Workload Modeling

This section reviews published studies related to multimodal fusion approaches in cognitive workload modeling. The aim is to summarize the main findings related to the application of multimodal fusion in classification of mental states and cognitive workload in particular.

Nowadays, in many fields, information about a given phenomenon can be obtained using different types of sources (sensors). Joint analysis of this data promises a more comprehensive and informative view of the problem at hand and alleviates the limitations encountered when depending on a single data source (i.e., noise or missing data). Early research in multimodal data fusion is traced back to the first half of the 20th century [61]. Since then, it has attracted further research attention [62]–[64]. Using data from multiple independent sensors allows systems to be more robust and resilient against failure than a single sensor, because information from multiple sensors is exploited to provide a better outcome [60].

Although multimodal data fusion approaches have been investigated in several studies in the literature for mental state classification in general and cognitive workload in particular, most of these studies did not report significant

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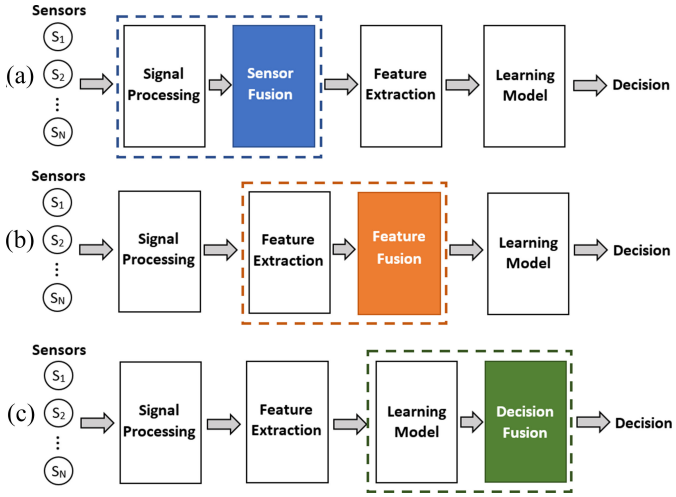


Fig. 2. Different fusion levels. (a) Fusion of sensor data (in blue). (b) Sensor data represented by feature vectors and its corresponding fusion (in orange). (c) Decision fusion after classification models (in green).

performance improvement of the fusion models over single-modality models.

A. Systematic Review Methodology

A literature review about multimodal data fusion for workload assessment was performed using Scopus and Google Scholar. This literature review includes only those studies that were related to the assessment of workload using more than one sensor modality (e.g., EEG, heart rate, and pupillometry). In addition, to be included in our review, a study needed to report the workload assessment results. The search terms are as follows: (multi AND modal) AND (data OR sensors OR measures OR biosignal) AND (fusion OR combination OR concurrent) AND (cognitive OR mental) AND (workload OR load OR state, OR processing). This initial search produced 394 papers.

Then, the literature review was refined by adding filters on the electronic databases. We included studies published since 2000. The search was limited to only papers written in English. We scanned journals, book chapters, and conference proceedings, and excluded work-in-progress papers, theses, and books. The search was also limited to Computer Sciences, Engineering, and Neuroscience fields. After applying these filters, the search results were reduced to 128 papers.

Finally, we evaluated the appropriateness of the collected papers to determine whether the articles met the inclusion criteria. The abstract of the identified studies was examined for content accuracy. Studies that were inconsistent with our search criteria, repeated studies from the same authors, or studies that did not report on numerical results of workload classification/recognition accuracy were excluded. This final step identified 34 papers, which are used for our literature review.

B. EEG and fNIRs-Based Approaches

EEG and fNIRs can serve as complementary modality to each other. EEG has high temporal resolution but lacks spatial specificity, while fNIRs, on the other hand, are spatially specific yet lack the time precision due to the delay in the hemodynamic response. Fusion of EEG and fNIRs has been investigated in different studies for mental state classification in brain–computer interface applications. Leamy *et al.* [65] combined features extracted from seven EEG and fNIRs channels into one classification model to differentiate two mental states (motor imagery and rest). Although combining features from multiple modalities showed higher classification accuracies, there was no statistical significance test to verify the results. Fazil *et al.* [66] used an LDA metaclassifier to combine decisions from an EEG-based classifier and an fNIRs-based classifier to classify two mental states in both subject-dependent and subject-independent scenarios. Different EEG features were used in subject-dependent scenarios to alleviate subject variability (i.e., PSD features were extracted from alpha band or beta band based on correlation with class label in the training data). Results showed statistically significant improvement when using the fusion approach over either EEG or fNIRs-based classification. Yin *et al.* [67] extracted 672 EEG time–phase–frequency features and 1728 fNIRs features. Different combinations of features were used to build classification models for the classification of a motor imagery task, a 1%–5% improvement in classification accuracy was achieved using fused features. However, no statistical analysis was reported for obtained results.

Tomita *et al.* [68] used EEG and fNIRs multimodalities in a cooperative way to improve steady-state visual evoked potentials (SSVEPs) classification. fNIRs-based classifier is used to detect the ON and OFF state while EEG-based classifier is used to classify eight different commands. Higher classification accuracy was obtained using the hybrid approach. However, no statistical tests were performed to verify the results.

Herff *et al.* [69] reported that feature-level fusion of the two modalities increased the robustness of classification using 66 features extracted from 3 EEG channels and 47 features extracted from 28 fNIRs sources. However, no statistical tests were performed to measure significance of results. Feature-level fusion improved performance over EEG alone by 3% in discriminating low versus high workload. However, there was no difference in performance when the workload levels increased from two to five.

Blokland *et al.* [70] combined fNIRs and EEG into decision-level fusion system to classify subject movement. Adding fNIRs significantly improved the classification accuracy for different classes. In the area of cognitive workload classification, few studies have investigated the feasibility of combining EEG and fNIRs in improving the workload classification task. Coffey *et al.* [71] investigated the combination of EEG and fNIRs to classify workload. Independent classifiers were built for each sensor and the classification results were fused to produce the final decision. Overall results did not show significant improvement of the fusion over an EEG-based model. Putze *et al.* [72] performed fusion of EEG and fNIRs at the decision level using metaclassifier to discriminate audio versus visual perception. For fNIRs, oxyhemoglobin (HbO) and deoxyhemoglobin (HbR) features were extracted from 170 fNIRs channels. For EEG, PSD and ERP features were extracted from 10 EEG channels resulting in total 38 EEG

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features. The metalevel fusion approach significantly outperformed classifiers based on single modality (EEG or fNIRs) in classifying mental states in both subject-dependent and subject-independent scenarios.

Liu *et al.* [73] proposed an approach to combine EEG and fNIRs at the decision fusion level by taking into consideration the different temporal resolutions of the two modalities. A classifier was trained to classify EEG PSD features at single-stimulus level (2.5 s epoch). The output scores from the 16 stimuli (of a block of 40 s) were averaged to produce the final EEG score. Another classifier was trained to classify fNIRs features extracted from each block (40 s epoch) to produce the fNIRs score. A metaclassifier was then trained to combine the EEG score and the fNIRs score for final classification. Results, however, did not show significant improvement in performance when using the fusion approach over either EEG alone or fNIRs alone. Liu *et al.* [74] applied decision-level fusion using different combinations of EEG, fNIIRs, and physiological sensors. Results showed that combining EEG and fNIRs statistically outperformed other decision fusion and single modality-based models.

C. EEG and Eye Tracking-Based Approaches

Fernandez and Dünser [3] combined six EEG-based features with average pupil diameter together into one classification model to classify cognitive workload. Their results showed that the fusion model significantly outperformed single-modality models in classifying two-level workload. Borys *et al.* [75] investigated several combinations of EEG and pupillometry features using different classification algorithms to classify mental workload states. Results showed that models based on eye tracking features alone achieved higher accuracy than fusion-based models. Brouwer *et al.* [21] suggested that EEG and eye pupillometry can complement each other to derive higher level understanding that neither of them can achieve separately. In their study, cognitive workload was assessed while participants were performing monitoring task. Eye tracking features were shown to better distinguish between hits and misses while EEG features were shown to better distinguish between targets and nontargets. So, combining features from both modalities into one model increased total accuracy. However, no statistical tests were reported in the study. Zheng *et al.* [25] compared two fusion strategies: feature-level fusion, and multimodal deep learning to combined EEG and eye tracking features to classify four emotions (happy, sad, fear, and neutral). In the multimodal deep learning strategy, two different neural networks were trained separately using the two modalities. Two hidden layers of EEG and eye networks were further concatenated as the input of the upper neural network. Results showed that both fusion approaches significantly outperformed single modality-based classification. In particular, the deep fusion model outperformed the feature-level fusion mode.

Other studies [76], [77] performed a joint analysis on EEG and eye pupillometry-based features to discover correlations with different workload states. However, the authors in these studies did not report on the differences among single-modality and multimodalities classification.

D. EEG and ECG-Based Approaches

Wilson and Russell [19] combined features from EEG and ECG sensors into one classification model to classify cognitive workload level. However, the authors did not report on the relative contribution of the different sensors to classification performance. Chandra *et al.* [78] used EEG and ECG recordings to measure cognitive load during a multiattribute task battery (MATB-II) before and after meditation. The classification task aimed to identify low-level and high-level workload. For the EEG data, PSD features, root mean squared (RMS), and engagement index were used; while for the ECG data, the R-R mean, the sympathovagal balance index (SVI), and the number of braces of continuous R-R intervals known as pNN50. The data fusion was carried out in a single classifier. The results showed that engagement index was the most prominent features based on the ANOVA analysis. However, the authors did not report on the results of the classification using a single modality. Zhang *et al.* [79] combined EEG using PSD and ERP features, and ECG features to classify multilevel cognitive workload. A weight was calculated for each feature based on mutual information to be used in an attempt to enhance fusion performance. However, fusion models did not show significant performance over single-modality models (EEG PSD, EEG ERP, and ECG). Feature weight-based model showed significant improvement over other feature concatenation models (i.e., simple concatenation of features).

E. More Than Two Modalities

Chanel *et al.* [18] combined features from EEG, skin conductance, heart rate, BP, respiration, and temperature together into one classification model to classify mental states. The authors analyzed the relative contribution of each physiological signal on classifying mental states. However, results showed that combining multiple features did not significantly improve the performance over using EEG features alone.

Khalili and Moradi [80] extracted and combined 384 features for EEG, 14 features for respiratory, 4 features for skin conductance, and 4 features for peripheral sensors from five subjects into one model for classifying three emotion states. Although there was a feature selection step prior to classification, the authors did not mention the final number of selected features in their model. Results showed that classification using multimodal features did not improve classification performance over either EEG alone or peripheral alone.

Chanel *et al.* [81] studied emotions classification. Classification decisions were made by two different EEG-based classifiers and one classifier based on skin conductance, heart rate, BP, and respiration. The decisions from the three classifiers were then combined together. The adoption of three different classifiers increased the classification accuracy by 5%–9% when compared with single classifier based on EEG. However, no statistical test was used to measure the significance. Fusion of the different modalities at the feature level was performed where no improvement in accuracy

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was found. Putze *et al.* [82] used a simple majority voting fusion to combine multiple modalities together (skin conductance, pulse, respiration, and EEG) to classify cognitive workload in visual and cognitive tasks. The study showed the performance of decision-level fusion varies significantly from one task to another. For example, decision-level fusion was the best in one task while it was outperformed in other tasks. Christensen *et al.* [83] combined 189 features from EEG/EOG with another four features (cardiac interbeat intervals, blink rate, blink amplitude, and blink duration) into a single classification model to assess workload in simulated remote piloting. They also did not extensively report on the relative contribution of these variables, but mentioned that when classification was performed on the basis of EEG only, classification accuracy hardly decreased (about 2%). Hussain *et al.* [84] used a hybrid multimodal fusion model for cognitive workload classification. Features from physiological sensors (ECG, GSR, RESP, and Eye) were first combined into one classification model and task performance features were used in a separate classification model where subdecisions were then combined using majority voting. The hybrid fusion approach improved classification accuracy by 6% and that improvement was statistically significant. The same authors in a subsequent study [85] investigated the impact of affective interference that emerge due to the nature of the task or personal feelings (i.e., emotional arousal, frustration, and confusion) on the performance of single models (based on single modality) and hybrid fusion model of multiple modalities for cognitive workload classification. Three main modalities were considered in that study: 1) face modality; 2) physio modality; and 3) performance modality. All features from video chromatic and geometric channels (115 features) were considered as face modality, all features from physiological channels (9 eye tracking features, 84 ECG features, 67 RESP features, and 21 skin conductance features) were considered as physio modality, and performance features (reaction time and accuracy score) were considered as performance modality. Liu *et al.* [73] proposed, an approach to combine EEG and fNIRs at the decision fusion level by taking into consideration the different temporal resolutions of the two modalities. A classifier was trained to classify EEG PSD features at single-stimure fusion level, all features from different modalities were fed into three different classifiers and the decisions were fused together to calculate the final decision. Fusion of multiple modalities showed higher performance in some cases, however, no statistical tests were performed to measure significance.

Chen *et al.* [86] fused features extracted from speech, GSR, and performance measures (hand writing pen input) to classify three levels of cognitive workload. The classification results based on all three modalities improved the classification accuracy by 13.4% over speech and the pen features only.

Son *et al.* [87] investigated the effect of different feature combinations on driver workload recognition performance. Skin conductance, heart rate, eye movement data, and driving performance measures were combined in different models and their performances were evaluated and compared. Results showed that combining heart rate, eye movement, and driving performance measures achieved the highest classification performance among all models, including single and multiple modalities, however, no statistical significance was reported in the study. Wei *et al.* [88] combined features from three modalities into one classification model to recognize two cognitive workload levels (easy versus difficult). The modalities used in this article were EEG, eye tracking, and ECG. Model based on the combination of all modalities was shown to have the highest classification accuracy among all other multimodality and single-modality models. However, no statistical test was performed to validate the obtained results.

Hogervorst *et al.* [20] performed extensive studies to analyze the impact of different types of fusions on the performance of workload classification. In their study, the performance of different combinations of features were examined: single-feature classifier, where a single feature from one sensor (i.e., only alpha activity, theta activity or ERP from EEG, or heart rate from ECG) is used. Sensor-level fusion-based classifier, where multiple features from a single sensor are used (i.e., alpha and theta activities from EEG). Multisensor-based fusion classifier, where multiple features from more than one sensor are used (i.e., alpha activity from EEG sensor and heart rate from ECG sensor). For the last category, two fusion models were used: 1) feature-level fusion where all features are combined and fed into one classifier and 2) decision-level fusion where a classifier is built for each different sensor. Different classification models were compared in terms of workload classification accuracy. Results showed that EEG-based classifiers outperform other physiological sensors (i.e., ECG and skin conductance). A single feature-based EEG classifier (one that use only alpha activity or ERP component) provides the best result across all models. It was also reported that there is no significant differences between feature-level and decision-level fusion, with the fNIRs channels. Zhang *et al.* [89] performed extensive study to analyze the impact of different types of fusions on the performance of workload classification. In their study, feature-level fusion, decision-level fusion, and hybrid fusion (both feature level and decision level) were implemented using EEG, ECG, EMG, eye tracking, GSR, PPR sensors, and performance metrics. Hybrid fusion was performed where features from EEG and eye tracking were processed using one classification model to make one subdecision; while other modalities features were preprocessed separately to make other subdecisions. However, no thorough statistical analysis was performed to analyze the different fusion schemes. Although multimodal fusion increased detection accuracy, this increase was not statistically significant. Saitis and Kalimeri [90] showed that feature-level fusion statistically outperformed single modality classifier in estimating the cognitive workload. In their study, they combined features from EEG, EDA, and BVP sensors to recognize eight different environments. Kindsvater *et al.* [91] investigated three different decision fusion strategies (Markov fusion network, Kalman filter-based fusion, and weighted majority voting) to improve cognitive workload classification using multimodal data collected using Kinect, EMG, EDA, ECG, and Audio. Although

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TABLE I
MULTIMODAL DATA FUSION STRATEGIES APPLIED TO COGNITIVE WORKLOAD MODELING. MODELS WHICH SIGNIFICANTLY IMPROVES WORKLOAD CLASSIFICATION ARE MARKED WITH *

Feature level fusion
Chanel et al. [18]; Khalili and Moradi [80]; Chanel et al. [81]; Christensen et al. [83]; Chen et al. [86]; Son et al. [87] Wei et al. [88]; Hogervorst et al. [20]; Zhang et al. [89]; Saitis and Kalimeri [90]; Zhang et al. [79]; Leamy et al. [65]; Herff et al. [69]; Rozado and Dunser [3] *; Borys et al. [75]; Plechawska-Wk and Borys [76]; Lobo et al. [77]; Wilson et al.[19]; Chandra al. [78]; Yin et al. [67]; Zheng et al. [25] *
Decision level fusion
Chanel et al. [81]; Hogervorst et al. [20]; Zhang et al. [89]; Putze et al. [72] *; Fazil et al. [66] *; Blokland et al. [70] *; Liu et al. [73]; Putze et al. [82]; Coffey et al. [71]; Tomita et al. [68]; Brouwer et al. [21]; Liu et al. [74]
Hybrid
Hussain et al. [84] *; Hussain et al. [85]; Zhang et al. [89]; Saitis and Kalimeri [90] *

multimodal fusion increased classification accuracy, no statistical test was performed to measure the significance of this increase.

F. Discussion

Table I categorizes the studies covered in this article by the level of fusion used. Studies show that reported significant performance improvement of fusions systems over single modality are marked with "*" symbol. Most of the studies fall under the complementary fusion category where multiple measures are collected from different sensors and combined together to classify mental state. However, few studies investigated other fusion paradigms.

Fig. 3(a) and (b) summarizes the studies by the level of fusion and modalities, respectively. Feature-level fusion has attracted the attention in several studies with 60% of covered papers investigated different feature-level fusion approaches in their studies; almost double the number of papers adopting decision-level fusion (32%). Hybrid-level fusion is under-investigated (8%), probably due to the higher complexity accompanied with hybrid models. In terms of modalities, the majority of studies (49%) combined more than two modalities together. EEG was the common factor in 89% of the covered studies. In particular, 27% of the studies investigated the combination of EEG and fNIRs, while 16% of the studies combined EEG and Pupillometry. Fusion of EEG and ECG attracted 8% of the studies. To have a better understanding of which fusion level was most applied on different sensor combinations, Table II summarizes the covered studies by both the level of fusion applied and sensor combination used. An important observation is that most of the studies that combined EEG and fNIRs adopted decision-level fusion (70%), while the remaining combinations used feature-level fusion.

As shown in Table I, the success rate of applying multimodal fusion systems in workload assessment is low. Only 24.32% (9 out of 37) of the covered studies reported statistically significant improvement over single modality in assessing mental states. To better understand the best performing fusion strategy for different sensor combinations, Fig. 4 summarizes the success rate of the application of different

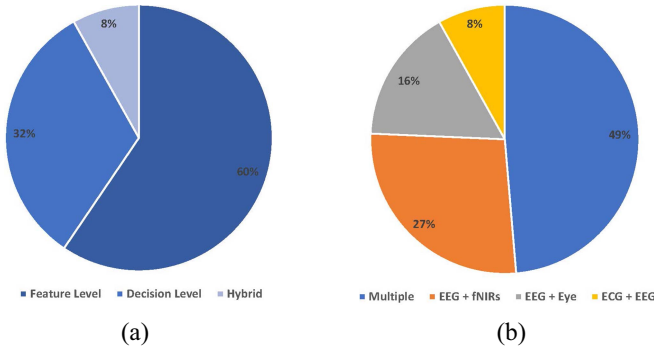


Fig. 3. Summary of publications. (a) Studies by fusion strategy. (b) Studies by modalities.

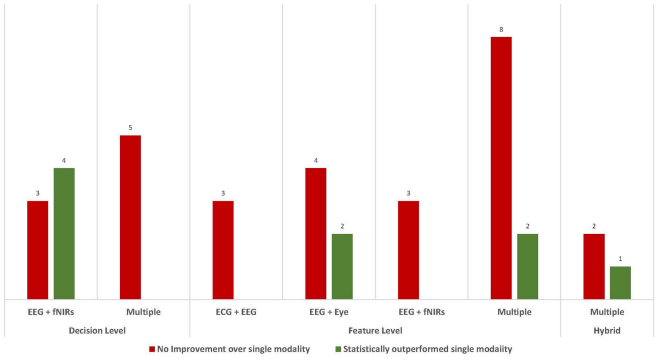


Fig. 4. Performance comparison with single modality.

fusion strategies for each combination of sensors. Combining EEG and fNIRs seems the most successful approach in the literature where four out of ten studies reported significantly better results; all these studies considered fusion at the decision level. The reason for the successful decision fusion of EEG and fNIRs could be attributed to the complementary nature of both sensors. EEG and fNIRs collect complementary measurements from the brain. EEG sensors have high temporal resolution, while fNIRs have better spatial resolution. Combining EEG with pupillometry comes next where two out of six studies reported statistically significant improvement over single modality; the two studies considered fusion of EEG and pupillometry at the feature level. Fusion of more than two modalities has the lowest success rate where only 3 out of 18 studies reported significant results; two of these considered feature-level fusion while the third study applied hybrid approach where both feature and decision levels were applied on different modalities. Combining more than two modalities showed the lowest success rate with 17% (3 out of 18 studies). Two of these studies considered feature-level fusion alone while the third one applied both feature and decision-level fusion. On the other hand, fusion of EEG and ECG at the feature level did not improve the performance in any study conducted in the literature.

To have a better understanding on how feature-level fusion performed in different studies, Table III summarizes the models' parameters for studies that conducted feature-level fusion. For each study, the number of features used from each sensor, the total number of features, the number of subjects in the

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TABLE II
FUSION APPROACHES WITH REPORTED SIGNIFICANT IMPROVEMENT GROUPED BY SENSORS USED IN THIS ARTICLE

Sensors used	Feature level fusion	Decision level fusion	Hybrid level fusion	Total #studies	Statistically better results	Fusion level with better results
EEG + ECG	3	0	0	**3**	**0**	-
Multiple sensors	10	5	3	**18**	**3**	Hybrid level, feature level
EEG + Eye	6	0	0	**6**	2	Feature level
EEG + fNIRs	3	7	0	**10**	**4**	Decision Level
Total	**22**	**12**	**3**	**37**	**9**	

study, and percentage of performance improvement achieved by fusion model over the best single modality are reported. It is worth noting that some of the studies covered in this article have less than ten subjects, which could affect the validity of results reported. These studies are highlighted in Table III using "+." The low success rate in these studies can be attributed to a number of issues associated with feature-level fusion implementation:

Unbalanced Feature Vector Lengths: Table III shows that features extracted from raw sensory data vary in length from one sensor to another. For example, EEG can be used to extract hundreds of features captured by power values in different frequencies at different locations on the scalp. On the other hand, other physiological features, such as pupil dilation and HRV, are represented by a small number of features (i.e., average). Simple concatenation of small feature vectors (i.e., ECG) with large feature vector (i.e., EEG) can lead to a smaller or insignificant contributions of these features.

Feature Noncommensurability: Different sensors are sensitive to different physical phenomena. As a result, the data is represented by heterogeneous measurement units with different scales. For instance, EEG measures the propagation of electrical fields generated by neurological activity, while fMRI measures changes in oxygenated blood level. Moreover, different measures reflect on different aspects of mental states that may or may not always correlate with cognitive workload. As a result, combining these measures could lead to biased or insignificant added value. For instance, HRV can indicate mental stress but can also indicate physical activity. Therefore, combining HRV with other measures may not be beneficial in all tasks as it could add useful information in some, while acting as noise in others.

Data Temporal Alignment: Different sensors have different temporal resolutions. For instance, while EEG offers high temporal resolution, other measurements, such as pupillometry, have a lower temporal resolution. Combining measurements with different temporal resolution could lead to noninstantaneous coupling even if the measurements are perfectly synchronized. Moreover, mental workload has a number of attributes which explains the mental resource demand/supply using time. These are instantaneous load, peak load, accumulated load, average load, and overall load [92]. Instantaneous load is the load at a given moment in time, which fluctuates during the performance of a task. Peak load reflects the maximum value of instantaneous load reached over certain time window. Accumulated load is the total amount of load experienced. Average load is the mean intensity of load over certain time window. Combining measures that represent different types of load could lead to contradictory results or on the best case small added value.

Fusion by Workload Task: To better understand how multimodal fusion performed for different types of workload, the task workload used in each study was extracted and performance of different modality combinations is evaluated for each category of these tasks as shown in Fig. 5. It is clear that the performance of multimodal fusion varies with task workload. While different modality combinations work better for some tasks than others, there are other tasks where fusion models did not show any improvement at all. For example, combining EEG with fNIRs is shown to work better in memory load and pyschomotor tasks while combination of EEG and pupillometry is shown to work better for mental arithmetic and emotion discrimination tasks. On the other hand, multimodal approach did not show improvement in visual workload, multitasking, or cognitive verbal transformation tasks. Six studies conducted in the literature used different modality combinations and fusion strategies to estimate visual workload showed that multiple modality did not improve performance over using a single sensor. The tasks used in these studies range from simple visual attention to virtual driving tasks and flight simulations. Two studies used verbal transformation as a secondary task to measure workload changes in dual task environments. Three other studies used fusion approaches to assess workload in multitasking environments using multiple attribute task battery (MATB) [93]. A common factor among most of these studies is that their workload tasks impose multiple cognitive requirements on human operators (i.e., cognitive, visual, and psychomotor) while fusion was applied at the feature level, which may be the reason why multimodal fusion has not improved performance in these tasks. Features extracted from some sensors can be used as an indicator for a particular type of workload, compiling these features together into a single model may not be helpful in formulating a general indicator of the cognitive workload.

A number of design considerations will be discussed in the next section that can improve the effectiveness of utilizing the multimodal sensory data in mental state recognition.

V. DESIGN CONSIDERATIONS FOR EFFECTIVE COGNITIVE WORKLOAD ESTIMATION

The choice of suitable fusion methods to estimate the cognitive workload depends on the context within which the data were collected. Several factors, such as the task design, type of workload measured, quality of data acquisition sensors, and

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TABLE III
MODEL PARAMETERS FOR FEATURE-BASED FUSION MODELS. "NS" IS USED WHERE PARAMETERS WERE NOT FOUND IN THE ORIGINAL STUDIES. "*" IS USED TO MARK STUDIES THAT REPORTED SIGNIFICANT CLASSIFICATION ACCURACY IMPROVEMENT FOR FUSION SYSTEMS. "+" IS USED TO MARK STUDIES WITH SMALL NUMBER OF SUBJECTS (LESS THAN 10). FIRST NINE COLUMNS SHOW THE NUMBER OF FEATURES USED FROM A GIVEN SENSOR. #TOT FEAT IS THE TOTAL NUMBER OF FEATURES USED. #SUB IS THE NUMBER OF SUBJECTS PER STUDY, AND IMPROVEMENT IS THE PERCENTAGE OF IMPROVEMENT BY FUSION OVER THE BEST INDIVIDUAL MODALITY

Reference	EEG	fNIRS	ECG	Eye	RESP	SC	Video	Peripheral	Performance	#Tot Feat	#sub	Improvement
Hussain et al. [85]	-	-	84	9	67	21	115	-	2	298	20	(-3 to 9)
Christensen et al. [83] +	189	-	1	3	-	-	-	-	-	193	8	NS
Zhang et al. [79]	2416	-	103	-	-	-	-	-	-	2519	10	NS
Borys et al. [75]	20	-	-	28	-	-	-	-	-	48	20	NS
Chanel et al. [18] +	6	-	2	-	4	4	-	8	-	24	4	-1
Wilson et al. [19]	25	-	12	-	-	-	-	-	-	37	10	NS
Chanel et al. [81]	18720	-	3	-	10	4	-	1	-	18738	11	(3 to 7)
Hussain et al. [84]	-	-	84	9	67	21	-	-	1	182	20	0.83
Hogervorst et al. [20]	1207	-	4	3	2	2	-	-	-	1218	14	(3 to 9)
Zhang et al. [89]	42	-	9	6	-	-	-	-	4	61	10	0.99
Lobo et al. [77]	16	-	-	5	-	-	-	-	-	21	20	NS
Plechawska-Wojcik [76] +	1	-	-	7	-	-	-	-	-	8	3	NS
Leamy et al. [65] +	14	14	-	-	-	-	-	-	-	28	2	(5 to 6)
Yin et al. [67] +	672	1728	-	-	-	-	-	-	-	2400	6	(1 to 5)
Herff et al. [69]	66	47	-	-	-	-	-	-	-	113	10	(0 to 3)
Chandra et al. [78]	NS	-	NS	-	-	-	-	-	-	NS	NS	(-20 to 0)
Khalili and Moradi [80] +	384	-	-	-	14	4	-	4	-	406	5	9
Son et al. [87]	-	-	1	2	-	1	-	-	2	6	30	19.23
Wei et al. [88]	1	-	1	1	-	-	-	-	-	3	14	(-9 to 4)
Chen et al. [86]	-	-	-	-	-	1	1	-	7	9	12	(1.4 to 15.1)
Rozado and Dunser [3] *	6	-	-	1	-	-	-	-	-	7	23	8
Zheng et al. [25] *	280	-	-	31	-	-	-	-	-	311	15	5.55

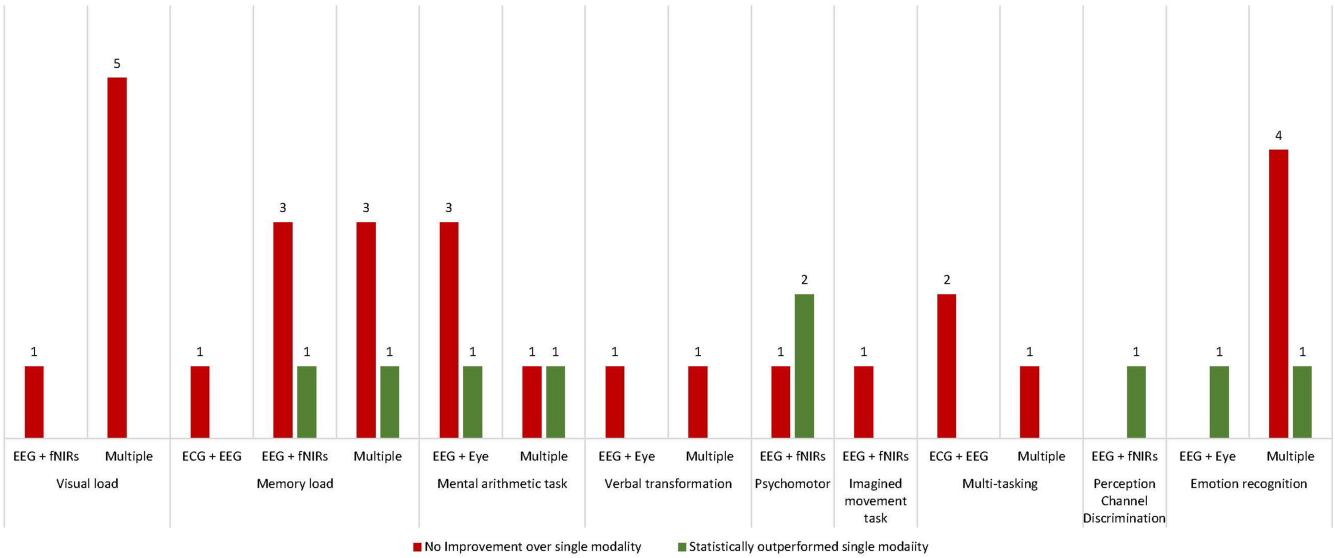


Fig. 5. Performance comparison across workload task.

confounding factors in the environment, determine the most suitable fusion strategy. For example, in tasks performed in outdoor environments, a potentially superior fusion needs to be achieved using a decision-level fusion of EEG and eye tracking. While confounding factors, such as light and body movement, could affect the quality of each modality, if these factors were independent, decision fusion could reduce the bias introduced by each confounding factor. For example, illumination could impact eye tracking, while body movements could impact EEG. Assuming no correlation between illumination and body movements, a decision fusion could overcome the noise caused by these factors.

Recently, there has been a growing interest in adaptive automation systems (also known as human autonomy teaming) that can adapt to human operator's changing requirements in real time [23], [94]–[97]. A system, where a robot team-mate recognizes human team-mate cognitive states (i.e., workload, stress, and fatigue), enables the robot to select appropriate behaviors to support its human team-mate. Human autonomy teaming provides flexibility so that the operator is more involved when the workload is manageable. When the operator is overloaded, on the other hand, the robot could prioritize its tasks to alleviate the workload. The efficient workload estimation models are, therefore,

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necessary to build such adaptive human autonomy teaming systems.

Cognitive workload is a complex phenomenon that requires careful design in terms of how, when, and which modalities can be fused. To effectively estimate the cognitive workload, a multimodal fusion system should meet a number of characteristics as follows.

Robustness: Review of the literature has shown that a single workload measure is not reliable to capture workload changes in all tasks all the time. A measure may respond to a particular task while it may not respond to other types of tasks [98]. Moreover, studies suggest that there is a considerable variability among individuals in the responses to workload [54]. The ability of human operators to respond to task demands depend on a number of factors, including general background, personality, motivation, attentiveness, trust, etc.

1) *Sensor Validation:* To handle data imperfection there should be a mechanism to alleviate the effects of confounding factors on the quality of sensor data (i.e., signal noise and artifacts).
2) *Metric Redundancy:* More than one measure of the same aspect of workload is necessary to address sensor failures and also to accommodate individual variability in workload responses for different sensors (motivation, task importance, attitude, etc.).
3) *Trust Mechanism:* Fusion of multiple measures of the same aspect should be based on the trust of fusion module in low-level sensors.

Diversity: Diversity is the key to data fusion if implemented at the appropriate level. Single-modality measures reflect different views of workload and respond differently to changes in workload. For example, measures that are linked to metabolic responses reflect global state of workload and respond more slowly to changes in workload state (i.e., cerebral blood flow velocity). Other measures reflect certain aspects of workload (i.e., pupil dilation was shown to reflect attention and engagement [99]) and respond to changes in workload more immediately [97], [100]. Combining measures that are not all related with the same rather with various aspects of workload can better complement each other and enhance workload classification.

Sensor Interaction: The relationship between different modalities can be dynamic and change overtime. Sometimes, different modalities can provide complementary measures to each other while at other times one modality is used to constrain another [68]. A fusion architecture should be able to recognize such interactions and adapt its fusion strategies according to the relationship dynamics.

Conflict Resolution Mechanism: A fusion strategy that can handle very different decisions from different sensors is necessary. This calls for context-aware fusion, especially, in real-time systems whereby the resolution strategy could leverage the information embedded in the context to disambiguate the situation. The research reviewed in this article relied on pieces of work done in an experimentally controlled environment, where the context is defined with an experimental protocol. In a real-world setting where context could vary, the task complexity component in Fig. 1 along with a context-detection module become necessary to offer the fusion system with information to disambiguate a situation and resolve a conflict among the different models used by the system.

VI. Conclusion

Multimodal data fusion is a well-established research area that has shown success in different domains. In cognitive state modeling domain there is a wide literature addressing multimodal fusion at different levels. However, the community was lacking a comprehensive and systematic review of the multimodal fusion in the research area of cognitive workload modeling. This article aimed at providing a systematic categorization and review of the multimodal fusion approaches in the cognitive workload modeling. The systematic review attempted to unlock the potential of multimodal data fusion in modeling cognitive workload; which affords unprecedented opportunities to further deepen our understanding of the workload based on various data measures. The review also covered data fusion in relevant domains such as emotion recognition. Limitations of using a single modality for workload modeling and the benefits of using multiple sensors to assess workload were discussed which give the motivations and advantages of multimodal fusion in this regard.

Despite the potential benefits of multimodal fusion and massive work that has been done in the cognitive workload literature, exploiting the additional diversity that multiple sensors offer is still at its very preliminary stages. The challenges affecting the fusion design choices were discussed. Distinctive design considerations were then introduced. These considerations are raised by the emergence of adaptive automation systems that require new research to adapt current state-of-the-art approaches and techniques of multimodal fusion. The most promising venues for the future lie in developing better models that can exploit the richness of data while also consider the confounding factors affecting the data quality in real time.

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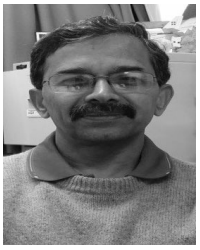
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